TEXORA · LECTURE SHEET

Process Control in Blow Room

Blow room control keeps three quantities in balance: cleaning efficiency high, total waste low, and fibre damage minimal. This sheet covers the routine measurements — trash analysis of feed and lap, waste audit, lap weight control — and how beater settings and grid settings are adjusted from them.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 3 · SPECIALIZATION — YARN MANUFACTURING

1. The three controlled quantities

The blowroom is judged on a triangle of measures that pull against each other. **Cleaning efficiency** (trash removed as a percentage of trash in) should be high. **Total waste %** should be low, because every point of waste is cotton bought and not spun. **Fibre damage** — shortening and nep creation by the beaters — should be minimal, though it is the hardest of the three to measure directly. Any single measure can be improved by wrecking another: wide-open grids raise cleaning and waste together; violent beating raises cleaning and damage together. Control means holding all three inside their norms simultaneously, for the mixing actually running (Topic 2 gives its expected trash, 3.74% in the example recipe).

2. Measuring cleaning efficiency

Trash is measured on feed and lap samples with a trash analyser (a small opening-and-collecting machine that separates trash from a weighed sample). Cleaning efficiency uses trash masses, exactly as in YM-3207 Topic 1:

$$\text{CE (\%)} = (\text{trash in feed} - \text{trash in delivery}) / \text{trash in feed} \times 100$$

WORKED EXAMPLE — BLOWROOM PERFORMANCE AUDIT

A line processes 500 kg of the Topic 2 mixing in a shift. Feed trash is measured at 6.0% (a dirtier lot than the recipe average — the reason for the audit). The line delivers 457.5 kg of lap; lap trash tests 1.9%.

Trash in = 6.0% of 500 = 30.0 kg.

Lap out = 457.5 kg, so total waste = 500 − 457.5 = 42.5 kg = **8.5%**.

Trash in lap = 1.9% of 457.5 = 8.69 kg.

Trash removed = 30.0 − 8.69 = 21.31 kg → CE = 21.31/30.0 × 100 = **71.0%**.
Of the 42.5 kg waste, 21.3 kg is trash; the remaining 21.2 kg is fibre and invisible loss. Waste analysis of the droppings found 7.3% of feed as visible waste, leaving 8.5 − 7.3 = **1.2% invisible loss** — inside the usual 1–2% band, so no leak or weighing fault is suspected.
Verdict: CE 71% is normal, waste 8.5% is acceptable for a 6%-trash feed, and invisible loss is normal — the line is in control; the higher waste follows the dirtier feed, not a machine fault.

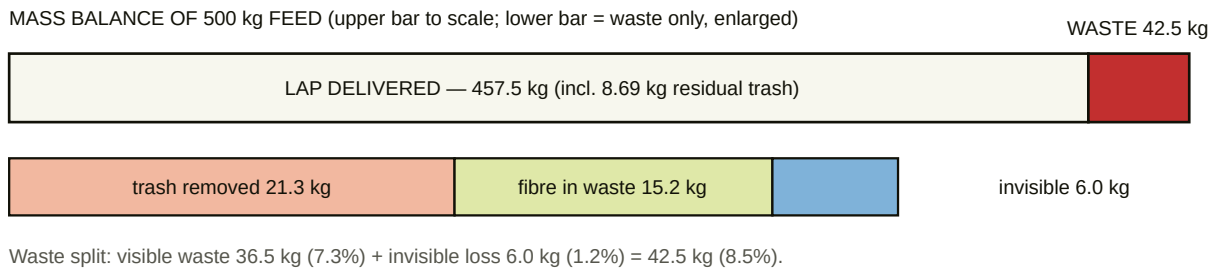


Figure 1 — The worked example as a mass balance. The lap bar and waste bar sum to the 500 kg feed; the lower bar splits the waste into trash removed, fibre lost and invisible loss (6.0 kg = 1.2% of feed).

3. Settings control

CONTROL	EFFECT OF INCREASING	WATCH QUANTITY
Beater speed	More opening and cleaning; more fibre rupture and neps	Lap nep level, waste %
Grid-bar opening (angle/gap)	More trash and more good fibre escape	Waste %, droppings fibre content
Beater-to-grid distance	Closer: more aggressive cleaning, more damage	CE, fibre length in lap
Feed rate (line loading)	Higher: tufts larger, cleaning per pass falls	Lap trash %, unopened tufts
Fan / air flow at cages	Carries tufts on; too high pulls fibre into waste ducts	Filter waste amount

The rule of adjustment is one change at a time, re-measured after a full lap cycle. Settings are recorded per mixing: a setting sheet that cleans a 3.7%-trash mix will over-clean or waste fibre on a 2% mix and under-clean a 6% mix like the audit example.

4. Lap weight control

Lap weight (g/m) and its CV% are checked by weighing complete laps and cut metre lengths. The scutcher's regulator should hold lap CV near or below about 1%; a rising CV with correct average

points to irregular feed from the last opener, while a shifted average points to the regulator setting. Because card sliver weight derives directly from lap weight (YM-3207 Topic 2), lap control is the first line of count control for the whole mill.

5. Routine control schedule

- **Each shift:** lap weight and CV sample; waste bins weighed against feed (waste %).
- **Daily:** trash test on feed mixing sample and on lap; droppings fibre check (a handful test backed by periodic analysis).
- **Weekly:** cleaning efficiency calculation on the mass basis; beater and grid setting verification against the setting sheet; nep level in lap where equipment allows.
- **On any mixing change:** repeat the full audit — the norms belong to the mixing, not to the machines alone.

Glossary

Trash analyser — laboratory machine that opens a weighed sample and collects its trash for weighing. **Droppings** — waste falling under beaters and grids. **Invisible loss** — waste found only by difference between feed and all weighed outputs. **Setting sheet** — the recorded standard settings for a machine for a given material. **Regulator** — the scutcher mechanism that varies feed roller speed with sheet thickness to hold lap weight constant. **Tuft size** — the average mass of opened fibre clusters leaving a machine; smaller tufts clean better.

Key points

• Control holds cleaning efficiency, waste % and fibre damage together; each can be "improved" by damaging another.
• Audit example: CE 71.0%, waste 8.5%, invisible loss 1.2% on a 6%-trash feed — in control.
• Invisible loss of 1–2% is normal; far above that suggests leaks, air-flow carryover or weighing errors.
• Beater speed, grid opening and feed rate are the working adjustments — changed one at a time and re-measured.
• Lap weight CV near 1% is the blowroom's evenness duty; it feeds directly into card sliver and yarn count control.

Practice

1. A line feeds 800 kg at 4.2% trash and delivers 752 kg of lap at 1.4% trash. Calculate cleaning efficiency and waste %.

2. Droppings analysis shows visible waste of 9.6% while total waste measures 10.1%. What is the invisible loss, and is it a concern?
3. After a change to a dirtier mixing, waste rises from 7% to 9% but cleaning efficiency is unchanged. Explain why this can be correct operation rather than a fault, and what extra check you would make before accepting it.

Answer key

1. Trash in = 0.042 × 800 = 33.6 kg. Trash in lap = 0.014 × 752 = 10.53 kg. Removed = 23.07 kg. CE = 23.07/33.6 × 100 = **68.7%**. Waste = (800 − 752)/800 × 100 = **6.0%**.
2. Invisible loss = 10.1 − 9.6 = **0.5%** — below the usual 1–2% band and not a concern; it is the total waste level itself that deserves attention in this case.
3. A dirtier feed contains more trash to remove, so more mass must leave as waste to reach the same lap cleanliness; with CE unchanged the line is doing the same job on harder material. Extra check: analyse the droppings for good-fibre content — if the extra waste is mostly fibre rather than trash, the grids are too open and it is a setting fault after all.
